

TROITSK, Russian Federation

WMO: 287480

Lat: 54.083N

Lon: 61.617E

Elev: 192

StdP: 99.04

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.0	-27.9	-34.1	0.2	-30.9	-31.0	0.2	-27.7	10.3	-5.9	9.8	-6.1	1.0	270	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	31.9	18.9	30.0	18.4	28.3	17.9	20.7	28.5	19.9	27.4	19.0	26.0	3.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.3	23.6	17.3	12.6	22.6	16.4	12.0	22.1	60.5	28.6	57.7	27.3	54.8	26.0	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.4	7.1	-34.8	34.7	3.4	2.0	-37.3	36.1	-39.3	37.3	-41.2	38.5	-43.7	39.9		
(5)			-34.9	22.1	3.4	1.4	-37.4	23.1	-39.3	23.9	-41.2	24.7	-43.7	25.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.5	-15.3	-13.8	-5.9	5.5	13.5	18.6	19.8	18.3	12.4	4.6	-5.0	-12.2	(6)
(7)		DBStd	13.90	6.65	6.80	5.51	5.66	4.95	4.48	3.90	4.28	4.56	4.89	6.95	7.07	(7)
(8)		HDD10.0	3464	784	666	493	156	23	2	0	1	24	178	451	687	(8)
(9)		HDD18.3	5636	1042	899	752	384	163	51	31	55	187	427	701	946	(9)
(10)		CDD10.0	1078	0	0	0	22	132	259	303	257	96	9	0	0	(10)
(11)		CDD18.3	207	0	0	0	0	13	58	76	53	7	0	0	0	(11)
(12)		CDH23.3	2291	0	0	0	10	213	652	780	542	94	0	0	0	(12)
(13)		CDH26.7	772	0	0	0	1	50	231	263	197	30	0	0	0	(13)
(14)	Wind	WSAvg	2.9	2.4	2.7	3.2	3.4	3.4	3.1	2.6	2.5	2.7	3.0	2.9	2.7	(14)
(15)	Precipitation	PrecAvg	378	19	17	21	24	37	48	61	48	26	30	24	22	(15)
(16)		PrecMax	487	39	48	49	62	88	107	129	165	56	97	52	56	(16)
(17)		PrecMin	220	1	3	2	0	1	12	16	9	2	2	3	5	(17)
(18)		PrecStd	82	11	10	14	16	22	28	30	32	16	21	15	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.2	2.2	8.9	24.8	30.2	33.6	34.6	33.7	30.6	20.7	10.1	2.7	(19)
(20)			MCWB	0.1	0.7	5.3	13.0	15.8	18.8	19.8	19.7	15.8	10.7	7.1	0.8	(20)
(21)		2%	DB	-1.3	0.4	5.0	20.8	28.1	31.4	31.4	31.2	25.7	17.6	6.9	1.0	(21)
(22)			MCWB	-2.1	-1.0	2.1	11.3	15.4	18.2	19.3	19.2	14.6	10.1	4.7	-0.3	(22)
(23)		5%	DB	-3.2	-1.6	3.0	17.9	25.9	29.4	29.4	28.9	22.8	14.9	5.0	-0.6	(23)
(24)			MCWB	-4.1	-2.6	1.1	10.0	14.5	17.6	19.3	18.5	14.2	8.6	3.0	-1.7	(24)
(25)		10%	DB	-4.8	-3.6	1.6	15.0	23.2	27.2	27.7	26.4	20.2	12.2	3.3	-2.4	(25)
(26)			MCWB	-5.6	-4.6	0.2	8.6	13.2	17.2	18.8	17.7	13.2	7.6	1.7	-3.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	1.0	5.9	13.7	18.1	21.0	22.0	21.5	17.6	12.7	7.2	1.2	(27)
(28)			MCDWB	1.1	2.2	8.9	22.2	26.0	30.2	28.9	29.8	27.5	18.0	8.8	2.5	(28)
(29)		2%	WB	-2.0	-1.0	2.5	12.0	16.5	19.7	21.0	20.2	15.8	10.9	5.0	-0.1	(29)
(30)			MCDWB	-1.2	0.3	4.4	19.3	25.3	27.8	28.4	28.6	22.9	16.3	6.8	0.9	(30)
(31)		5%	WB	-4.0	-2.6	1.3	10.5	15.3	18.7	20.2	19.4	14.6	9.3	3.3	-1.7	(31)
(32)			MCDWB	-3.3	-1.6	2.6	17.1	23.7	26.5	27.3	26.9	21.1	13.9	4.7	-0.6	(32)
(33)		10%	WB	-5.6	-4.6	0.3	8.9	14.1	17.8	19.4	18.3	13.5	7.9	1.8	-3.3	(33)
(34)			MCDWB	-4.8	-3.6	1.6	14.7	21.8	25.5	26.0	24.6	19.4	11.7	3.2	-2.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.1	9.5	9.0	10.1	12.0	11.6	10.6	10.5	10.6	8.8	7.1	7.6	(35)
(36)			MCDBR	8.0	8.0	8.5	15.1	16.1	14.6	13.5	14.3	14.6	14.0	7.9	7.1	(36)
(37)			MCWBR	7.6	7.3	6.5	7.5	6.5	5.3	4.7	5.0	6.4	7.6	6.0	6.5	(37)
(38)		5% WB	MCDWR	7.8	7.9	7.7	13.8	14.1	12.8	11.8	12.5	12.6	12.4	7.2	7.0	(38)
(39)			MCWBR	7.4	7.3	6.1	7.3	6.4	5.4	4.8	5.0	6.1	7.3	5.8	6.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.285	0.316	0.370	0.412	0.409	0.406	0.427	0.444	0.386	0.355	0.315	0.285	(40)	
(41)		taud	2.080	2.040	1.972	2.119	2.228	2.312	2.242	2.169	2.285	2.266	2.192	2.067	(41)	
(42)		Ebn at Noon	637	745	781	809	837	844	815	771	773	704	601	544	(42)	
(43)		Edh at Noon	70	102	138	137	130	121	128	130	101	82	63	58	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.02	2.12	3.66	4.57	5.45	5.95	5.68	4.56	3.20	1.71	0.88	0.65	(44)	
(45)		RadStd	0.07	0.10	0.27	0.47	0.51	0.49	0.39	0.45	0.25	0.27	0.09	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page